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A Rational Sextic Associated with a Desargues Configuration

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Abstract. In this paper we consider a Desargues configuration in the projective plane, i.e. ten points and ten lines, on each line we have three of the points and through each point we have three of the lines. We construct a rational curve of order 6 which has a node at each of the ten points. We have never seen this kind of curve in the literature, but it is well known that for any n there exists a rational curve of order n which has $[(n-1)(n-2)]/2$ nodes and if $n = 6$ we find a sextic with ten nodes. The purpose of this paper is to obtain a sextic of this kind as a locus of points in connection with special projectivities of the plane associated with the Desargues configuration and to find a rational parametric representation of it. A large part of this paper is done with MACSYMA: it is an application of computer algebra in algebraic geometry. Special cases, where we find a quintic, a quartic or a cubic, are given in the last section.

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1. Projectivities Associated with a Desargues Configuration

First we work in the complex projective plane, denoted by $\mathcal{P}$. Actually, the results remain true in a projective plane over any algebraically closed field. In Section 4 we will work in a real complexified plane.

Let us consider a Desargues configuration ($\mathcal{DC}$ for short) in $\mathcal{P}$: ten mutually different points and ten mutually different lines, such that we have on each of the lines exactly three of the points and through each of the points exactly three of the lines. Special cases where we have on some lines four of the points are treated in the last section.

The notations are as follows: ABC and $A'B'C'$ are two perspective triangles, the lines $AA' = a''$, $BB' = b''$, $CC' = c''$ are concurrent at a point S and the points $AB \cap A'B' = c \cap c' = C''$, $BC \cap B'C' = a \cap a' = A''$, $CA \cap C'A' = b \cap b' = B''$ are collinear on a line s.

It is well known that a triangle in $\mathcal{P}$ and its polar triangle with regard to a non-degenerate conic $\mathcal{K}$ in $\mathcal{P}$ are in perspective, and conversely, if you give a $\mathcal{DC}$, then there exists exactly one non-degenerate conic $\mathcal{K}$ such that the complete $\mathcal{DC}$ is autodual with respect to $\mathcal{K}$, i.e. each point of the $\mathcal{DC}$ is the pole of a line of the $\mathcal{DC}$ and each line of the $\mathcal{DC}$ is the polar line with regard to $\mathcal{K}$ of a point of the $\mathcal{DC}$. In the following, we call $\mathcal{K}$ the *conic* of the $\mathcal{DC}$.

Next, recall that there are six kinds of projectivities in the projective plane $\mathcal{P}$:

1. The identical transformation of $\mathcal{P}$.

2. The homologies: there is a fixed point S and a line s, each point of which is invariant.
3. The special homologies: homologies with $S \in s$.
4. Projectivities with exactly one fixed point S and one fixed line s, with $S \in s$.
5. Projectivities with two fixed points $S_1,\ S_2$ and two fixed lines $s_1,\ s_2$; in this case on one fixed line there are two fixed points and on the other fixed line there is one fixed point. For instance: $S_1,\ S_2 \in s_1$ and $S_1 = s_1 \cap s_2$.
6. There are three invariant points $S_1,\ S_2,\ S_3$, which are the vertices of a triangle and the sides $S_1S_2 = s_3,\ S_2S_3 = s_1,\ S_3S_1 = s_2$ are the invariant lines.

A projectivity f of the projective plane $\mathcal{P}$ is completely determined if we give four points $A_1,\ A_2,\ A_3,\ A_4$ and the four images $f(A_1),\ f(A_2),\ f(A_3),\ f(A_4)$, so that no three of the points $A_1,\ A_2,\ A_3,\ A_4$ (and $f(A_1),\ f(A_2),\ f(A_3),\ f(A_4)$, respectively) are collinear.

In connection with our $\mathcal{DC}$, with points $A,\ B,\ C,\ A',\ B',\ C',\ A'',\ B'',\ C'',\ S$, we consider the projectivity which is defined as follows: $A \to A',\ B \to B',\ C \to C', X \to X$, where the fixed point X is a given point of $\mathcal{P}$. We assume throughout this paper that X is not a point of the $\mathcal{DC}$, and not a point on a line of the $\mathcal{DC}$. We say that such a point is *outside* the $\mathcal{DC}$. It is clear that for a given point X, *outside* the $\mathcal{DC}$, the projectivity is always well determined since no three of the four points $A,\ B,\ C,\ X$ (and $A',\ B',\ C',\ X$, respectively) can be collinear and the projectivity will be in general of the sixth kind: we will have three invariant points $X,\ Y,\ Z$ which are the vertices of a triangle. But for special points X of $\mathcal{P}$, the projectivity will be of the fifth kind and for yet more special points, the projectivity is of the fourth kind. The principal aim of this paper is to find the locus of the points X in $\mathcal{P}$ such that the projectivity f, determined by $f(A) = A',\ f(B) = B',\ f(C) = C',\ f(X) = X$, has two invariant points and two invariant lines, i.e. such that f is of the fifth kind. And from this it is not difficult to find the finite number of points X such that f has only one fixed point or is of the fourth kind.

2. Two Lemmas

LEMMA 1. *Consider the $\mathcal{DC}$ in $\mathcal{P}$ with points $A,\ B,\ C,\ A',\ B',\ C',\ A'',\ B'',\ C''$, S. Determine a projectivity f of $\mathcal{P}$ as follows: $f(A) = A',\ f(B) = B',\ f(C) = C'$ and $f(X) = X$, where X is a point* outside *the $\mathcal{DC}$. Then, if Y is any invariant point of f, the polar line y of Y with regard to the* conic $\mathcal{K}$ *of the $\mathcal{DC}$ is an invariant line of f. In other words, the invariant configuration of f is self-dual with respect to $\mathcal{K}$.*

Proof. First we prove that if the fixed point X is *outside* the $\mathcal{DC}$, any other invariant point Y of f (if there is such other fixed point) is also *outside* the $\mathcal{DC}$. Suppose that Y is a point of s, different from $A'',\ B'',\ C''$. Then f is a homology with axis s and center S, which is impossible since X is *outside* the $\mathcal{DC}$. Next, suppose that Y is on one of the lines $a,\ b,\ c,\ a',\ b',\ c'$, for instance on $a = BC$. Since $f(BC) = B'C'$, we have $Y = BC \cap B'C' = A''$. Put $a'' \cap a = A_1$ and $a'' \cap a' = A'_1$. We get the following equality of cross-ratios: $(A''\ C\ A_1\ B) = (A''\ C'\ A'_1\ B')$ and thus $f(A_1) = A'_1$, which means that the line AA' is invariant. This is impossible, because in that case the point $XA'' \cap AA'$ would be invariant too. Finally, suppose that Y is on one of the lines a'', b'' or c'', for instance on a''. Then the line a'' is invariant and it is easy to prove that A'' would be a fixed point, which is impossible.

So, any fixed point Y is *outside* the $\mathcal{DC}$. Consider the polar line y of Y with regard to the *conic* $\mathcal{K}$. Clearly $y \neq AB = c$: put $y \cap c = A_1$, $CY \cap c = C_1$, $y \cap c' = A'_1$ and $C'Y \cap c' = C'_1$. Since $f(CY) = C'Y$ and $f(c) = c'$, we have $f(C_1) = C'_1$. The polar lines of the points B', A', A'_1, C'_1 with regard to $\mathcal{K}$ are $CA = b$, $CB = a$, CY, CA_1, respectively. From this we get the following equality of cross-ratios $(A\ B\ C_1\ A_1) = (B'\ A'\ A'_1\ C'_1) = (A'\ B'\ C'_1\ A'_1)$, and since $f(A) = A'$, $f(B) = B'$, $f(C_1) = C'_1$, we find $f(A_1) = A'_1$ or $f(y \cap c) = y \cap c'$. In the same way, we have for the intersections of y with $BC = a$ that $f(y \cap a) = y \cap a'$ and thus $f(y) = y$.

Next, if f fixes a line, then it fixes the pole of the line with respect to $\mathcal{K}$. This follows from a similar reasoning on the map induced by f on the set of lines of the plane. This completes the proof.

Because of the foregoing, we know that the following cases will occur (remark that f cannot be an homology, since X is *outside* the $\mathcal{DC}$):

(a) The projectivity f has three fixed points X, Y, Z which are the vertices of a polar triangle with respect to $\mathcal{K}$.
(b) We have two invariant points X and Y. In this case the line XY is a tangent line of $\mathcal{K}$ with X or Y on $\mathcal{K}$. If for instance X is a point of $\mathcal{K}$, then the polar line y of Y (which is not a point of $\mathcal{K}$) with regard to $\mathcal{K}$ and the line XY are the fixed lines of f.
(c) The projectivity f has X as only fixed point: then X is necessarily a point of $\mathcal{K}$ and the tangent line of $\mathcal{K}$ at X is the unique invariant line of f.

Next, we introduce the following notation: the conic through five points A_1, A_2, A_3, A_4, A_5 is denoted by $\mathcal{C}(A_1, A_2, A_3, A_4, A_5)$.

In the $\mathcal{DC}$ with points A, B, C, A', B', C', A'', B'', C'', S there are ten pairs of perspective triangles. Each point of the $\mathcal{DC}$ is a perspective center for two perspective triangles. For instance the triangles $A''C''B$ and $C'A'S$ are perspective with center B'. In the statement of Lemma 1, the corresponding points of the projectivity f are the corresponding points of the perspective triangles ABC and $A'B'C'$ and $f(X) = X$. We have:

LEMMA 2. *For any of the* 20 *projectivities which have as corresponding points the corresponding points of two perspective triangles of the $\mathcal{DC}$ and for which a given point X*, outside *the $\mathcal{DC}$, is a fixed point, the invariant points (and lines) are always* *the same. Moreover, any invariant point of these projectivities is a common point of the five conics* $\mathcal{C}(A, B, C, S, X)$, $\mathcal{C}(A', B', C', S, X)$, $\mathcal{C}(B'', A'', C, C', X)$, $\mathcal{C}(C'', B'', A, A', X)$ *and* $\mathcal{C}(A'', C'', B, B', X)$.

Proof. The point C'' is the perspective center of the triangles $AA'B''$ and $BB'A''$, with corresponding points (A, B), (A', B'), (B'', A''), which means that the lines AB, $A'B'$, $B''A''$ are concurrent with common point C''. Consider the projectivity h determined by $h(A) = B$, $h(A') = B'$, $h(B'') = A''$, $h(X) = X$. The lines through A are mapped on the lines through B and it is well known (theorem of Steiner) that the locus of the intersection point of corresponding lines of these pencils is a conic. In this way corresponds with the pair $(A, h(A) = B)$ the conic $\mathcal{C}(A, B, S, C, X)$, with the pair $(A', h(A') = B')$ the conic $\mathcal{C}(A', B', S, C', X)$ and with the pair $(B'', h(B'') = A'')$ the conic $\mathcal{C}(B'', A'', C, C', X)$. Of course, any other invariant point of h, apart from X, will also be a common point of these three conics.

Next, consider the point A'' which is the perspective center of the triangles $CC'B''$ and $BB'C''$ and determine the projectivity g by $g(C) = B$, $g(C') = B'$, $g(B'') = C''$, $g(X) = X$. In the same way as before, we find that any fixed point of g is a common point of the conics $\mathcal{C}(C, B, S, A, X)$, $\mathcal{C}(C', B', S, A', X)$ and $\mathcal{C}(B'', C'', A, A', X)$.

The point B'' is the perspective center of the triangles $AA'C''$ and $CC'A''$. Again in the same way, we find that any invariant point of the projectivity k determined by $k(A) = C$, $k(A') = C'$, $k(C'') = A''$, $k(X) = X$, is a common point of the conics $\mathcal{C}(A, C, S, B, X)$, $\mathcal{C}(A', C', S, B', X)$ and $\mathcal{C}(C'', A'', B, B', X)$.

It follows that the three projectivities h, g and k have the same fixed points because these points are common points of the conics $\mathcal{C}(A, B, S, C, X)$ and $\mathcal{C}(A', B', S, C', X)$. Finally, consider any other of the 20 projectivities, for instance f determined by $f(A) = A'$, $f(B) = B'$, $f(C) = C'$, $f(X) = X$. The corresponding conics are now $\mathcal{C}(A, A', C'', B'', X)$, $\mathcal{C}(B, B', A'', C'', X)$, $\mathcal{C}(C, C', A'', B'', X)$ and their common points are the invariant points of f. But these three conics contain the common fixed points of h, g and k, and therefore f has the same invariant points. This completes the proof.

Remarks. 1. The five conics mentioned in Lemma 2 are always non-degenerate if X is a point *outside* the $\mathcal{DC}$. For instance, no three of the points A, B, S, C, X can be collinear and $\mathcal{C}(A, B, S, C, X)$ is non-degenerate.

2. Consider any of the 20 projectivities associated with the $\mathcal{DC}$ such as described above and call it f. We have the following cases:

(a) The projectivity f has three fixed points X, Y, Z, which are common points of the five conics $\mathcal{C}(A, B, C, S, X)$, $\mathcal{C}(A', B', C', S, X)$, $\mathcal{C}(B'', A'', C, C', X)$, $\mathcal{C}(C'', B'', A, A', X)$ and $\mathcal{C}(A'', C'', B, B', X)$.

(b) The projectivity f has two invariant points X, Y with for instance X on $\mathcal{K}$. The line XY is the tangent of $\mathcal{K}$ at X and Y is a point of the five conics, which have at X the same tangent line, the polar line of Y with regard to $\mathcal{K}$.

(c) The projectivity f has one fixed point X, a point of $\mathcal{K}$. The invariant line of f is the tangent x of $\mathcal{K}$ at X and the five conics through X have x as tangent line at X and are mutually osculating at X (three common points at X).

3. The Main Theorem

Consider again the $\mathcal{DC}$ with points $A,\ B,\ C,\ A',\ B',\ C',\ A'',\ B'',\ C'',\ S$. We still assume that on every line of the $\mathcal{DC}$ there are exactly three points of the configuration.

We look for the locus of the points P of the plane $\mathcal{P}$, *outside* the $\mathcal{DC}$, such that the projectivity f determined by $f(A) = A',\ f(B) = B',\ f(C) = C'$ and $f(P) = P$ (or any of the 20 projectivities which have the corresponding vertices of two perspective triangles on the $\mathcal{DC}$ as corresponding points and P as fixed point) has only two invariant points. Moreover, we will look for the finite number of points P, such that the projectivity has only one fixed point.

From the foregoing, we already know that if f has no more than two fixed points, the *conic* $\mathcal{K}$ of the $\mathcal{DC}$ is a component of this locus: the projectivity f always has one invariant point on $\mathcal{K}$. But if there is a second invariant point, this point will not be on $\mathcal{K}$; it is a point on the tangent line of $\mathcal{K}$ at the first fixed point and so, we have, besides $\mathcal{K}$, a second component of our locus.

THEOREM. *The locus of the points P of the plane $\mathcal{P}$,* outside *the $\mathcal{DC}$, such that the projectivity f given by $f(A) = A',\ f(B) = B',\ f(C) = C',\ f(P) = P$ has no more than two invariant points consists of two components: the first component is the* conic *$\mathcal{K}$ of the $\mathcal{DC}$ and the second component is a rational curve $\mathcal{S}$ of order 6 which has a node at each of the ten points of the $\mathcal{DC}$. The two curves $\mathcal{K}$ and $\mathcal{S}$ have six common points which are the points P such that f has only one fixed point.*

Note. Of course the ten nodes of the sextic and the 20 intersection points of the lines of the $\mathcal{DC}$ with the *conic* $\mathcal{K}$ of the $\mathcal{DC}$ are not *outside* the $\mathcal{DC}$. See Remark 1 after the proof.

Proof. We already know that $\mathcal{K}$ is a component of the locus. In order to find the second component, we will work as follows. If Q is a variable point of $\mathcal{K}$, then this second component is the locus of the second intersection point P of the conic $\mathcal{C}(S,\ A,\ B,\ C,\ Q)$ with the tangent line of $\mathcal{K}$ at the point Q. Actually, we may choose four other conics (see Lemma 2 above).

A purely geometrical (and complete) proof of the theorem is not trivial. It is possible to obtain a proof using algebraic $(m,\ n)$-relations, but since we are not a specialist in this area, we prefer to give a proof which is an example of an application of computer algebra in algebraic geometry. A rational parametric representation of the curve can be calculated without a computer, but we did it with MACSYMA. A short and nice form of the equation of the sextic (due to J. Bilo) is given in Remark 3 after the proof.

The coordinates system in $\mathcal{P}$ is chosen as follows. We do not start with the $\mathcal{DC}$ but with the *conic* $\mathcal{K}$ of the $\mathcal{DC}$ and we give $\mathcal{K}$ the equation $X^2 - YZ = 0$. This means that $(0,\ 1,\ 0)$ and $(0,\ 0,\ 1)$ are points on $\mathcal{K}$, with tangent lines $Z = 0$ and $Y = 0$, respectively, and that $(1,\ 1,\ 1)$ is on $\mathcal{K}$.

Next, without loss of generality, we give S the coordinates $(1,\ 0,\ 0)$ and we give the three lines of the $\mathcal{DC}$ through S the equations $Y - Z = 0$, $Y + bZ = 0$ and $cY + Z = 0$, on which we have the points $A,\ B, C$, respectively (and of course also $A',\ B',\ C'$, respectively). Finally, we give the point A the coordinates $(a,\ 1,\ 1)$. Now the $\mathcal{DC}$ is completely determined and from a straightforward calculation we get the following coordinates for the ten points and the following equations for the ten lines:

$$\begin{array}{lll} S(1,\ 0,\ 0), & A(a,\ 1,\ 1), & B(a(bc+1),\ b(c-1),\ 1-c), \\ C(a(bc+1),\ 1-b,\ c(b-1)), & & \\ A'((c-1)(b-1),\ 2a(bc+1),\ 2a(bc+1)), & & \\ B'(1-b,\ -2ab,\ 2a), & C'(c-1,\ -2a,\ 2ac), & A''(0,\ 1, -1), \\ B''(0,\ b,\ 1), & C''(0,\ 1,\ c). & \end{array}$$

Equations:

$$\begin{array}{ll} s\colon X = 0, & a\colon (c-1)(b-1)X - a(bc+1)Y - a(bc+1)Z = 0, \\ b\colon (1-b)X - aY + abZ = 0, & c\colon (c-1)X - acY + aZ = 0, \\ a'\colon 2aX - Y - Z = 0, & b'\colon 2a(bc+1)X + (c-1)Y + b(1-c)Z = 0, \\ c'\colon 2a(bc+1)X + c(1-b)Y + (b-1)Z = 0, & a''\colon Y - Z = 0, \\ b''\colon Y + bZ = 0, & c''\colon cY + Z = 0. \end{array}$$

Of course the values of $a,\ b,\ c$ are not arbitrary. We recall that we work with a $\mathcal{DC}$ of the most general kind: on each of the ten lines there are only three points of the $\mathcal{DC}$. The conditions, in order to have a $\mathcal{DC}$ and to have this general situation, are:

$$\begin{array}{lll} a \neq 0\ (A' \neq S), & a \neq 1\ (A \text{ not on } \mathcal{K}), & a \neq -1\ (A \text{ not on } \mathcal{K}), \\ b \neq 0\ (B'' \text{ not on } \mathcal{K}), & b \neq 1\ (C \neq S), & b \neq -1\ (B'' \neq A''), \\ c \neq 0\ (C'' \text{ not on } \mathcal{K}), & c \neq 1\ (B \neq S), & c \neq -1\ (C'' \neq A''), \\ bc + 1 \neq 0\ (A' \neq S), & bc - 1 \neq 0\ (B'' \neq C''), & \end{array}$$

$$a^2(bc+1)^2 + b(c-1)^2 \neq 0\ (B \text{ not on } \mathcal{K}),$$

$$a^2(bc+1)^2 + c(b-1)^2 \neq 0\ (C \text{ not on } \mathcal{K}),$$

$$(c-1)^2(b-1)^2 - 4a^2(bc+1)^2 \neq 0\ (A' \text{ not on } \mathcal{K}),$$

$$(b-1)^2 + 4a^2b \neq 0\ (B' \text{ not on } \mathcal{K}), \qquad (c-1)^2 + 4a^2c \neq 0 (C' \text{ not on } \mathcal{K}),$$

$$(c-1)(b-1) - 2a^2(bc+1) \neq 0\ (A \neq A').$$

Next, the easiest way seems to consider the pencil of conics through the points A'', B'', C, C'. The non-homogeneous equation of the pencil determined by A'', B'', C, C' is:

$$\alpha X(cY+1)+\beta(2aX-Y-1)((1-b)X-aY+ab)=0,$$

with homogeneous parameters α and β.

A variable point Q of $\mathcal{K}$ has non-homogeneous coordinates $(t,\ t^2)$, where t is a parameter, and we find the non-homogeneous equation of our locus by eliminating t from the equation of the conic $\mathcal{C}(A'',\ B'',\ C,\ C',\ Q)$ of the pencil through Q and the equation of the tangent line of $\mathcal{K}$ at Q:

$$\begin{aligned}&(2at-t^2-1)((1-b)t-at^2+ab)X(cY+1)-\\&\quad -t(ct^2+1)(2aX-Y-1)((1-b)X-aY+ab)=0 \qquad (1)\end{aligned}$$

$$2tX-Y-t^2=0. \qquad (2)$$

The elimination of t is almost impossible without a computer. We used MACSYMA and the result is of the following kind:

$$(Y-X^2)(\text{‘ a polynomial of order 6 in } X,\ Y\text{’})=0.$$

The first factor $Y-X^2$ gives of course the left side of the equation of $\mathcal{K}$, and the second factor determines the equation of the sextic. It is a very long equation, we will not give it. We only remark that there are no terms in X^6, neither in X^5 and the coefficient of X^4 is (use *factor(ratcoef(equation,* X^4)) in MACSYMA):

$$16a^2(cY+bc-c+1)(Y(bc-b+1)+b). \qquad (3)$$

Next, it is possible to obtain from (1) and (2) a rational parametric representation of the sextic. If we eliminate Y out of (1) and (2), then we find an expression in X and t, which is quadratic in X, but we know that this polynomial is divisible by $(X-t)$ and thus we can get X as a rational expression in t. Substituting it in (2), we find the rational expression of Y in t. The result is:

$$X=\frac{act^6+(abc-ac+a)t^4-(abc-ab+a)t^2-ab}{2act^5+(4abc-4ac+4a)t^3-(4a^2bc+2bc-2b-2c+4a^2+2)t^2+(2abc-2ab+2a)t}$$

$$Y=-\frac{(abc-ac+a)t^4-(2a^2bc+bc-b-c+2a^2+1)t^3+(2abc-2ab+2a)t^2+ab}{act^4+(2abc-2ac+2a)t^2-(2a^2bc+bc-c-b+2a^2+1)t+(abc-ab+a)}.$$

A homogeneous parametric representation in X, Y, Z is:

$$\begin{aligned}X=&\ act^6+a(bc-c+1)t^4-a(bc-b+1)t^2-ab\\Y=&\ -2((abc-ac+a)t^5-(2a^2bc+bc-b-c+2a^2+1)t^4+\\&+(2abc-2ab+2a)t^3+abt)\\Z=&\ 2(act^5+(2abc-2ac+2a)t^3-(2a^2bc+bc-c-b+2a^2+1)t^2+\\&+(abc-ab+a)t).\end{aligned}$$

Next, is this second component of the locus a sextic and has it a node at each of the ten points of the $\mathcal{DC}$? That it is a sextic follows from the parametric representation and also from its long equation. Moreover, we observed that in this non-homogeneous equation there are no terms in X^6, neither in X^5, which means that $S(1, 0, 0)$ is a singular point of the sextic and the tangent lines at S are given by the coefficient (3) of X^4. Is S always a node, or can S be a cusp, in other words, is it possible that the tangent lines at S coincide? A straightforward calculation shows that this is the case if $(bc+1)(b-1)(1-c)=0$. If $b=1$, then $C=S$; if $c=1$, then $B=S$; if $bc+1=0$ then $A'=S$. Thus, S always is a node: a singular point with two different tangents. But S is in no way an exceptional point of the $\mathcal{DC}$ and therefore also the nine other points of the $\mathcal{DC}$ are nodes of the sextic.

Finally, we look for the points such that the projectivity f has only one invariant point. It is obvious that these points must be the common points of $\mathcal{K}$ and the sextic $\mathcal{S}$. Using numerical examples in MACSYMA, we noticed that we always got only six common points: $\mathcal{K}$ and $\mathcal{S}$ always were tangent at six points. MACSYMA cannot calculate the common points in the general situation (i.e. if you do not give values at a, b, c). But if $\mathcal{K}$ and $\mathcal{S}$ always have only six intersection points, instead of the expected 12 common points, then, substituting the homogeneous parametric representation of $\mathcal{S}$ in the polynomial X^2-YZ one must find that this polynomial of degree 12 in t actually is the square of a polynomial of the degree 6. And within a few seconds MACSYMA gave us the following nice result:

$$(act^6+(3a+3abc-3ac)t^4+(2b-2bc-2-4a^2-4cba^2+2c)t^3+$$
$$+(3a+3abc-3ab)t^2+ab)^2.$$

So, $\mathcal{K}$ and $\mathcal{S}$ have six common points at which they have the same tangent line. And these are the points P of the plane $\mathcal{P}$ such that the projectivity f has only P as invariant point. This completes the proof.

Remarks. 1. The ten points of the $\mathcal{DC}$ or the ten nodes of the sextic are *particular points* of the locus, in the sense that they are not *outside* the $\mathcal{DC}$. We said that the invariant points of any projectivity f determined by the corresponding points of any pair of perspective triangles in the $\mathcal{DC}$ and by $f(X)=X$ are always the same. This is only true if X is a point *outside* the $\mathcal{DC}$, which was assumed. But then, why are the ten points of the $\mathcal{DC}$ points of the sextic $\mathcal{S}$?

Consider the projectivity determined by $f(A)=A'$, $f(B)=B'$, $f(C)=C'$, $f(X)=X$, where X is a given point. We said that we assume that X is a point *outside* the $\mathcal{DC}$. But in our construction of the sextic, we use the variable point $Q(t, t^2)$ of $\mathcal{K}$ and for instance for $t=0$, we find the point $Q(0, 0)$ which is a point of s, not *outside* the $\mathcal{DC}$. The projectivity f, where $X=(0, 0)$, is an homology with center S. So, S corresponds in this way with the intersection point $Q(0, 0)$ with $\mathcal{K}$ of the polar line s of S with regard to $\mathcal{K}$. On each line of the $\mathcal{DC}$ we have two common points with $\mathcal{K}$. And this is the reason why we find each of the ten points twice (a node) on the sextic: any point of the $\mathcal{DC}$ corresponds in this way with the two intersection points of its polar line with regard to $\mathcal{K}$ with $\mathcal{K}$ and the ten nodes of the sextic are the centers of the homologies which leave the $\mathcal{DC}$ invariant as a set.

2. The expression of X in the parametric representation of the sextic can be factorized as follows:

$$X = a(t^2 - 1)(t^2 + b)(ct^2 + 1).$$

From this, we see which values of t give the three nodes A'', B'', C'' on the line s with equation $X = 0$, for instance with $t = \pm 1$ corresponds the point $A''(0,\ 1,\ -1)$.

A new parametric representation of the sextic, where Y and Z (or Y', Z') have analogous expressions as X, is obtained by choosing (for instance) the points A, B'', C'' as basis points of a new reference system. The coordinate transformation is:

$$X = aX', \qquad Y = X' + bY' + Z', \qquad Z = X' + Y' + cZ'.$$

A straightforward (computer) calculation gives the following parametric representation in the new coordinates:

$$\begin{aligned}
X' &= (1 - bc)(t^2 - 1)(t^2 + b)(ct^2 + 1) \\
Y' &= (t^2 - 2at + 1)(ct^2 + 1)((c - 1)t^2 + 2a(bc + 1)t - b(c - 1)) \\
Z' &= (t^2 + b)(t^2 - 2at + 1)(c(b - 1)t^2 - 2a(bc + 1)t - b + 1).
\end{aligned}$$

The ten nodes of the sextic are now:

$$\begin{aligned}
&A(1,\ 0,\ 0), \qquad B''(0,\ 1,\ 0), \qquad C''(0,\ 0,\ 1), \\
&B(bc + 1,\ 0,\ -1 - b), \qquad C(bc + 1,\ -1 - c,\ 0), \\
&A''(0,\ 1 + c,\ -1 - b), \qquad S(bc - 1,\ 1 - c,\ 1 - b), \\
&A'((bc - 1)(c - 1)(b - 1),\ (c - 1)((1 - c)(b - 1) + 2a^2(bc + 1)), \\
&\quad (b - 1)((1 - b)(c - 1) + 2a^2(bc + 1))), \\
&B'((bc - 1)(1 - b),\ (1 - c)(1 - b) - 2a^2(bc + 1),\ (1 - b)^2 + 4a^2 b), \\
&C''((bc - 1)(c - 1),\ -(1 - c)^2 - 4a^2 c,\ (1 - b)(c - 1) + 2a^2(bc + 1)).
\end{aligned}$$

The equation of the first component of the locus, i.e. the *conic* $\mathcal{K}$ of the $\mathcal{DC}$, becomes:

$$\begin{aligned}
&(a^2 - 1)X'^2 - bY'^2 - cZ'^2 - (1 + b)X'Y' - (1 + c)X'Z' - \\
&\quad -(bc + 1)Y'Z' = 0.
\end{aligned}$$

3. In the foregoing remark the vertices of a triangle of the $\mathcal{DC}$ are the basis points of the reference system and in this connection J. Bilo obtained in the following

way a nice and short form for the equation of the sextic. The coordinates of the points of the $\mathcal{DC}$ are:

$$A(1,\ 0,\ 0),\quad B(0,\ 1,\ 0),\quad C(0,\ 0,\ 1),\quad S(1,\ 1,\ 1),$$
$$A'(1-d,\ 1,\ 1),\quad B'(1,\ 1-e,\ 1),\quad C'(1,\ 1,\ 1-f),$$
$$A''(0,\ e,\ -f),\quad B''(d,\ 0,\ -f),\quad C''(d,\ -e,\ 0),$$

with $def \neq 0$ and $def \neq ef + fd + de$. The line coordinates are:

$$BC(1,\ 0,\ 0),\quad CA(0,\ 1,\ 0),\quad AB(0,\ 0,\ 1),$$
$$B'C'(ef-e-f,\ f,\ e),\quad C'A'(f,\ fd-f-d,\ d),$$
$$A'B'(e,\ d,\ de-d-e),\quad SA(0,\ 1,\ -1),\quad SB(1,\ 0,\ -1),$$
$$SC(1,\ -1,\ 0),\quad s(d^{-1},\ e^{-1},\ f^{-1}).$$

The equation of the conic $\mathcal{K}$ of the $\mathcal{DC}$ is now:

$$\sum((ef-e-f)X^2 + 2dYZ) = 0,$$

where the sum is taken over the cyclic permutations of $X,\ Y,\ Z$ and $d,\ e,\ f$. The tangential equation of $\mathcal{K}$ is:

$$\sum((1-d)U^2 + 2VW) = 0.$$

A variable conic $\mathcal{C}$ of the pencil of conics through the points $S,\ A,\ B,\ C$ is given by: $\lambda YZ + \mu ZX + \nu XY = 0$ with $\lambda + \mu + \nu = 0$, or short $\Sigma\lambda YZ = 0$ with $\Sigma\lambda = 0(1)$.

For the tangential equation of this conic $\mathcal{C}$ we get:

$$\lambda^2U^2 + \mu^2V^2 + \nu^2W^2 - 2\mu\nu VW - 2\nu\lambda WU - 2\lambda\mu UV = 0$$

or

$$\sum(\lambda^2U^2 - 2\mu\nu VW) = 0(2).$$

Next, put the equation of $\mathcal{K}$ equal to $\Phi(X,\ Y,\ Z) = 0$ and express that the polar line of a point of the conic $\mathcal{C}$ with respect to $\mathcal{K}$ is a tangent line of $\mathcal{C}$: you get, from (1) and (2), the equation of the sextic in the following form

$$\sum\left(\lambda^2\left(\frac{\partial\Phi}{\partial X}\right)^2 - 2\mu\nu\frac{\partial\Phi}{\partial Y}\frac{\partial\Phi}{\partial Z}\right) = 0$$

with

$$\lambda : \mu : \nu = X(Y-Z) : Y(Z-X) : Z(X-Y).$$

Remark that you get also from this equation that the ten points of the $\mathcal{DC}$ are nodes of the sextic.

4. Next, recall the following formulas of Plucker (n is the degree of the curve, m is the class, i is the number of inflexions, d is the number of nodes, c is the number of cusps and b is the number of bitangents):

$$m = n(n-1) - 2d - 3c, \qquad i = 3n(n-2) - 6d - 8c,$$
$$n = m(m-1) - 2b - 3i.$$

For the sextic $\mathcal{S}$, we have $n = 6,\ d = 10,\ c = 0$ and thus $\mathcal{S}$ has class 10, has 12 inflexions and has 24 bitangents.

4. Working in a Real Complexified Projective Plane

Until now, we worked in a complex projective plane or in a projective plane over any algebraically closed field. In this section the plane $\mathcal{P}$ is real complexified, i.e. the coordinate system is real and points with real non-homogeneous coordinates (or homogeneous coordinates which are proportional with real homogeneous coordinates) are real. But we also consider imaginary points, i.e. points with non-homogeneous coordinates which are not both real or with homogeneous coordinates which are not proportional with real coordinates. Of course, the same holds for the lines in connection with their tangential coordinates.

We consider in $\mathcal{P}$ a real $\mathcal{DC}$: this means that each point and thus also each line of the $\mathcal{DC}$ is real. Such as in the foregoing sections, we still assume that we have on each line of the $\mathcal{DC}$ exactly three of the ten points.

Which possibilities have we for the sextic $\mathcal{S}$? That is, which points of the $\mathcal{DC}$ are crunodes (two real tangent lines) and which are acnodes (two conjugate imaginary tangents)? In order to answer this question, first we have to know which are the tangent lines at each of the ten nodes of the sextic.

LEMMA 3. *The point S of the $\mathcal{DC}$ is the perspective center for the triangles ABC and $A'B'C'$. Call S_1 and S_2 the two intersection points of the polar line $s = A''B''C''$ of S with regard to the* conic $\mathcal{K}$ *of the $\mathcal{DC}$. The conics $\mathcal{C}(S, A, B, C, S_1)$ and $\mathcal{C}(S, A', B', C', S_1)$ have the same tangent line at S and so do the conics $\mathcal{C}(S, A, B, C, S_2)$ and $\mathcal{C}(S, A', B', C', S_2)$. These two tangent lines are the tangents at the node S of the sextic $\mathcal{S}$. At any other of the ten nodes of $\mathcal{S}$, the tangent lines are constructed in the same way.*

Proof. We use the same coordinate system as in the proof of the main theorem:

$$S(1, 0, 0), \qquad A(a, 1, 1), \qquad B(a(bc+1), b(c-1), 1-c),$$
$$C(a(bc+1), 1-b, c(b-1)),$$
$$A'((c-1)(b-1), 2a(bc+1), 2a(bc+1)), \qquad B'(1-b, -2ab, 2a),$$
$$C'(c-1, -2a, 2ac), \qquad S_1(0, 0, 1), \qquad S_2(0, 1, 0).$$

A straightforward calculation gives us the following homogeneous equations:

$$\mathcal{C}(S,\ A,\ B,\ C,\ S_1):\ bXZ - a(bc+1)Y^2 + (bc-b+1)XY = 0.$$

$$\mathcal{C}(S,\ A',\ B',\ C',\ S_1):\ 2abXZ - (b-1)(c-1)Y^2 + \\ +2a(bc-b+1)XY = 0.$$

$$\mathcal{C}(S,\ A,\ B,\ C,\ S_2):\ a(bc+1)Z^2 + (c-bc-1)XZ - cXY = 0.$$

$$\mathcal{C}(S,\ A',\ B',\ C',\ S_2):\ (b-1)(c-1)Z^2 + 2a(c-bc-1)XZ - \\ -2acXY = 0.$$

Clearly the tangent line of $\mathcal{C}(S,\ A,\ B,\ C,\ S_1)$ and $\mathcal{C}(S,\ A',\ B',\ C',\ S_1)$ at S is $Y(bc-b+1)+bZ = 0$. And the tangent line of $\mathcal{C}(S,\ A,\ B,\ C,\ S_2)$ and $\mathcal{C}(S,\ A',\ B',\ C',\ S_2)$ at S is $cY + (bc-c+1)Z = 0$.

Next, we know from (3) in the proof of the main theorem that the non-homogeneous equation of the tangent lines at the node S of the sextic is given by:

$$16a^2(cY + bc - c + 1)(Y(bc-b+1)+b) = 0$$

and this completes the proof.

We assumed that on any line of the $\mathcal{DC}$ there are exactly three points of the $\mathcal{DC}$, which means that none of the ten points is on the *conic* $\mathcal{K}$ of the $\mathcal{DC}$. Consider again the point S. If the intersection points S_1 and S_2 of s with $\mathcal{K}$ are real, then the conics $\mathcal{C}(S,\ A,\ B, C,\ S_1)$ and $\mathcal{C}(S,\ A,\ B,\ C,\ S_2)$ are real and S will be a crunode. Remark that these conics cannot coincide. If S_1 and S_2 are conjugate imaginary, then these conics have an equation which is not real and they cannot have a real tangent line at S, otherwise these conics would be real. Actually, in this case the equations of $\mathcal{C}(S,\ A,\ B,\ C,\ S_1)$ and $\mathcal{C}(S,\ A,\ B,\ C,\ S_2)$ are conjugate imaginary and they have imaginary conjugate tangent lines at the real point S, i.e. S is an acnode. So, we have:

COROLLARY. *If the* conic $\mathcal{K}$ *of the real* $\mathcal{DC}$ *is imaginary, then each of the ten points of the* $\mathcal{DC}$ *is an acnode of the sextic* $\mathcal{S}$. *In this case the ten nodes are the only real points of* $\mathcal{S}$. *If* $\mathcal{K}$ *is real, then any point of the* $\mathcal{DC}$ *which is outside* $\mathcal{K}$, *i.e. which has real tangent lines to* $\mathcal{K}$, *is a crunode of* $\mathcal{S}$ *and any point of the* $\mathcal{DC}$ *which is inside* $\mathcal{K}$, *i.e. which has conjugate imaginary tangent lines to* $\mathcal{K}$, *is an acnode of* $\mathcal{S}$.

How many points of a real $\mathcal{DC}$, with a real *conic* $\mathcal{K}$, can be inside $\mathcal{K}$? An almost straightforward reasoning gives the following possibilities:

- One point of the $\mathcal{DC}$ is inside and the nine others outside.
- Two points, on a line of the $\mathcal{DC}$ (e.g. $A,\ A'$), are inside.
- Three collinear points (e.g. $A'',\ B'',\ C''$) are inside $\mathcal{K}$.
- Three points, which are the vertices of a triangle of the $\mathcal{DC}$ (e.g. $A,\ B,\ C$), are inside.

- Four points, the vertices of a triangle of the $\mathcal{DC}$ and its perspective centre (e.g. $A,\ B,\ C,\ S$), are inside $\mathcal{K}$.

Five or more points of the $\mathcal{DC}$ cannot be inside $\mathcal{K}$ (remark that for each point inside $\mathcal{K}$, there are three points, on the polar line of that point, outside $\mathcal{K}$). So, we find:

COROLLARY. *If the* conic $\mathcal{K}$ *of the real* $\mathcal{DC}$ *is real, the sextic has one, two, three or four acnodes (and nine, eight, seven or six crunodes).*

5. Special Desargues Configurations

In the foregoing, the $\mathcal{DC}$ was of the general kind, i.e. on each of the ten lines were exactly three of the ten points. In this section we consider the cases where we have on some line(s) four of the ten points. This will occur if one or two or three points of the $\mathcal{DC}$ are points of the *conic* $\mathcal{K}$ of the $\mathcal{DC}$.

The two lemmas of Section 2 still hold for these special cases and thus we can find the locus of the points P of $\mathcal{P}$ *outside* the $\mathcal{DC}$ such that the projectivity given by $f(A) = A',\ f(B) = B',\ f(C) = C',\ f(P) = P$ has no more than two invariant points in the same way as before.

The coordinates system and the parameters $a,\ b,\ c$ were chosen in the general situation so that:

(a) for $b = 0$, the point B'' becomes $(0,\ 0,\ 1)$, which is a point of $\mathcal{K}$ while b'' $(Y = 0)$ is the tangent line at B'' of $\mathcal{K}$ on which we have the four points $S,\ B'',\ B,\ B'$;

(b) for $b = c = 0$, the points $B'',\ C''$ become $(0,\ 0,\ 1),\ (0,\ 1,\ 0)$, respectively, which are two points of $\mathcal{K}$, with tangent lines b'' $(Y = 0)$ and c'' $(Z = 0)$, respectively, on which we find the points $S,\ B'',\ B,\ B'$ and $S,\ C'',\ C,\ C'$, respectively;

(c) for $b = c = 0,\ a = 1$, the triangle $AB''C''$ is inscribed in $\mathcal{K}$: on b'' $(Y = 0)$, we have the points $S,\ B'',\ B,\ B'$, on c'' $(Z = 0)$, we have the points $S,\ C'',\ C,\ C'$ and on $a'(2X - Y - Z = 0)$, we find the points $A,\ B',\ C',\ A''$.

5.1. ONE POINT OF THE $\mathcal{DC}$ IS A POINT OF $\mathcal{K}$

Put $b = 0$ in the $\mathcal{DC}$ which we used until now. We find the following coordinates for the ten points and the following equations for the ten lines:

$$S(1,\ 0,\ 0),\qquad A(a,\ 1,\ 1),\qquad B(a,\ 0,\ 1-c),\qquad C(a,\ 1,\ -c),$$
$$A'(1-c,\ 2a,\ 2a),\qquad B'(1,\ 0,\ 2a),\qquad C'(c-1,\ -2a,\ 2ac),$$
$$A''(0,\ 1,\ -1),\qquad B''(0,\ 0,\ 1),\qquad C''(0,\ 1,\ c).$$

Equations:

$$s\colon X = 0, \qquad a\colon (1-c)X - aY - aZ = 0, \qquad b\colon X - aY = 0,$$
$$c\colon (c-1)X - acY + aZ = 0, \qquad a'\colon 2aX - Y - Z = 0,$$
$$b'\colon 2aX + (c-1)Y = 0, \qquad c'\colon 2aX + cY - Z = 0,$$
$$a''\colon Y - Z = 0, \qquad b''\colon Y = 0, \qquad c''\colon cY + Z = 0.$$

On b'' we have the four points B'', B, S, B' and through B'' we have the four lines b'', b, s, b'. We work in exactly the same way as before, i.e. we eliminate t from equations (1), (2) in the proof of the main theorem (but with $b = 0$):

$$\begin{aligned}&(2at - t^2 - 1)(t - at^2)X(cY + 1)-\\ &\quad -t(ct^2 + 1)(2aX - Y - 1)(X - aY) = 0\\ &2tX - Y - t^2 = 0.\end{aligned} \tag{4}$$

The result is of the following kind:

$$Y(Y - X^2)(\text{'a polynomial of order five in } X,\ Y\text{'}) = 0.$$

The actual locus consists now, besides the *conic* $\mathcal{K}$, of a curve $\mathcal{Q}$ of order 5. The line $Y = 0$ or b'' corresponds with the value $t = 0$ (remark that (4) is divisible by t), and must be considered as a singular part of the locus. We will not give the equation of the quintic $\mathcal{Q}$, it is too long. But a rational parameter representation of $\mathcal{Q}$ is:

$$X = \frac{at(t^2 - 1)(ct^2 + 1)}{2act^4 + 4a(1-c)t^2 + 2(c - 2a^2 - 1)t + 2a},$$

$$Y = \frac{t^2(at - 1)(t(c-1) + 2a)}{act^4 + 2a(1-c)t^2 + (c - 2a^2 - 1)t + a}.$$

One can verify that the quintic contains the points S, B, B', B'' (and b'' is the tangent line of $\mathcal{Q}$ at B''), but these four points are not singular points of $\mathcal{Q}$. The $[(5-1)(5-2)]/2 = 6$ nodes of the quintic are the six other points A, A', A'', C, C', C'' of the $\mathcal{DC}$. Remark that if we take the line b'' into account as a part of the locus (besides $\mathcal{K}$), then this locus ($b'' + \mathcal{Q}$) still is a (now degenerate) sextic with the ten points of the $\mathcal{DC}$ as singular points.

Substituting a homogeneous parametric representation of $\mathcal{Q}$ in the homogeneous equation $X^2 - YZ = 0$ of $\mathcal{K}$, we find:

$$t^2(act^4 + 3a(1-c)t^2 + 2(c - 2a^2 - 1)t + 3a)^2 = 0.$$

The point B'' corresponds with $t = 0$. The points *outside* the $\mathcal{DC}$ which give projectivities of the fourth kind, are now four points of $\mathcal{K}$ where $\mathcal{K}$ and $\mathcal{Q}$ are tangent.

5.2. TWO POINTS OF THE $\mathcal{DC}$ ARE POINTS OF $\mathcal{K}$

Put $b = c = 0$ in the $\mathcal{DC}$. We find the following coordinates for the ten points and the following equations for the ten lines:

$$S(1,\ 0,\ 0),\quad A(a,\ 1,\ 1),\quad B(a,\ 0,\ 1),\quad C(a,\ 1,\ 0),$$
$$A'(1,\ 2a,\ 2a),\quad B'(1,\ 0,\ 2a),\quad C'(1,\ 2a,\ 0),\quad A''(0,\ 1,\ -1),$$
$$B''(0,\ 0,\ 1),\quad C''(0,\ 1,\ 0).$$

Equations:

$$s\colon X = 0,\quad a\colon X - aY - aZ = 0,\quad b\colon X - aY = 0,$$
$$c\colon -X + aZ = 0,\quad a'\colon 2aX - Y - Z = 0,\quad b'\colon 2aX - Y = 0,$$
$$c'\colon 2aX - Z = 0,\quad a''\colon Y - Z = 0,\quad b''\colon Y = 0,\quad c''\colon Z = 0.$$

On b'' we have the four points B'', B, S, B' and through B'' we have the four lines b'', b, s, b'. On c'' we have the four points C'', C, S, C' and through C'' we have the four lines c'', c, s, c'. We work again in the same way as before (but with $b = c = 0$) and we eliminate t from:

$$(2at - t^2 - 1)(t - at^2)X - t(2aX - Y - 1)(X - aY) = 0 \tag{5}$$

$$2tX - Y - t^2 = 0. \tag{6}$$

The result is:

$$\begin{aligned}&Y(Y - X^2)(a^2Y^3 + 2Y^2(4a^2X^2 - 4a^3X - 2aX + a^2)+\\ &\quad +Y(a^2 - 4aX - 8a^3X + 4X^2 + 16a^2X^2 + 16a^4X^2 - 8aX^3-\\ &\quad -16a^3X^3) + 16a^2X^4 - 16a^3X^3 - 8aX^3 + 8a^2X^2) = 0.\end{aligned}$$

The actual locus consists now of the *conic* $\mathcal{K}$ and of a curve $\mathcal{F}$ of order 4. A rational parameter representation of this quartic is:

$$X = \frac{at(t^2 - 1)}{2(2at - 1)(t - a)}\qquad Y = \frac{t^2(at - 1)(2a - t)}{(2at - 1)(t - a)}.$$

It is easy to verify that $\mathcal{F}$ contains the points B, B', B'', C, C', C'', which are regular points of $\mathcal{F}$ (b'' and c'' are the tangent lines of $\mathcal{F}$ at the points B'' and C'', respectively), and the $[(4 - 1)(4 - 2)]/2 = 3$ nodes of $\mathcal{F}$ are three other points of the $\mathcal{DC}$: A, A', A''.

The lines $Y = 0$ and $Z = 0$, or b'' and c'', are now singular parts of the locus. We lost $Z = 0$, because we worked in (5) and (6) with non-homogeneous coordinates $(X,\ Y)$ and with a non-homogeneous representation $(t,\ t^2)$ of $\mathcal{K}$. Remark that $b'' + c'' + \mathcal{F}$ still is a sextic with the ten points of the $\mathcal{DC}$ as singular points.

Substituting a homogeneous parametric representation of $\mathcal{F}$ in the homogeneous equation $X^2 - YZ = 0$ of $\mathcal{K}$, we find:

$$t^2(3at^2 - (4a^2 + 2)t + 3a)^2 = 0.$$

The point B'' corresponds with $t = 0$. The points *outside* the $\mathcal{DC}$ which give projectivities of the fourth kind, are now two points of $\mathcal{K}$ where $\mathcal{K}$ and $\mathcal{F}$ are tangent.

5.3. THREE POINTS OF THE $\mathcal{DC}$ ARE POINTS OF $\mathcal{K}$

Put $b = c = 0$ and $a = 1$ in the $\mathcal{DC}$. We find the following coordinates for the ten points and the following equations for the ten lines:

$$S(1, 0, 0), \quad A(1, 1, 1), \quad B(1, 0, 1), \quad C(1, 1, 0),$$
$$A'(1, 2, 2), \quad B'(1, 0, 2), \quad C'(1, 2, 0), \quad A''(0, 1, -1),$$
$$B''(0, 0, 1), \quad C''(0, 1, 0).$$

Equations:

$$s\colon X = 0, \quad a\colon X - Y - Z = 0, \quad b\colon X - Y = 0,$$
$$c\colon X - Z = 0, \quad a'\colon 2X - Y - Z = 0, \quad b'\colon 2X - Y = 0,$$
$$c'\colon 2X - Z = 0, \quad a''\colon Y - Z = 0, \quad b''\colon Y = 0, \quad c''\colon Z = 0.$$

On b'' we have the four points B'', B, S, B' and through B'' we have the four lines b'', b, s, b'. On c'' we have the four points C'', C, S, C' and through C'' we have the four lines c'', c, s, c'. On a' we have the four points A, B', C', A'' and through A we have the four lines a', b, c, a''. We work again in the same way as before (but with $b = c = 0$ and $a = 1$) and we eliminate t from:

$$(2t - t^2 - 1)(t - t^2)X - t(2X - Y - 1)(X - Y) = 0 \tag{7}$$

$$2tX - Y - t^2 = 0.$$

The result is:

$$Y(Y - X^2)(Y - 2X + 1)(Y^2 + 8X^2Y - 10XY + Y - 8X^3 + 8X^2) = 0.$$

The actual locus is now, besides $\mathcal{K}$, a cubic $\mathcal{T}$. A rational parametric representation of $\mathcal{T}$ is:

$$X = \frac{t(t+1)}{4t - 2} \qquad Y = \frac{t^2(2 - t)}{2t - 1}.$$

It is easy to see that A, B'', C'' are regular points of the cubic $\mathcal{T}$, with tangent lines a', b'', c'', respectively, that B, C, A'' are regular points of $\mathcal{T}$, and that the $[(3 - 1)(3 - 2)]/2 = 1$ node is the point A'.

The lines $Y = 0$, $Z = 0$ and $Y - 2X + 1 = 0$, or b'', c'' and a', are singular parts of the locus. The line b'' corresponds with the value $t = 0$ (remark that (7) is divisible by t), we lost the line c'' with equation $Z = 0$, because of the same reasons as in the foregoing section, and the line a' corresponds with the value $t = 1$. Remark that the degenerate curve $b'' + c'' + a' + \mathcal{T}$ still is a sextic with the ten points of the $\mathcal{DC}$ as singular points.

Since $\mathcal{K}$ and $\mathcal{T}$ are tangent at the three points B'', C'', A, there are no points *outside* the $\mathcal{DC}$ which give projectivities of the fourth kind.

Remarks. 1. The cubic $\mathcal{T}$ has a node and thus has three collinear inflexions. One can easily verify that these inflexions are the points B, C, A''.

2. The tangent lines of the cubic $\mathcal{T}$ at its node A' are constructed in the same way as in Lemma 3 of Section 4. The node A' is the perspective center of the triangles $AB''C''$ and $SC'B'$; if A'_1 and A'_2 are the intersection points of the polar line a of A' with regard to the *conic* $\mathcal{K}$, then $\mathcal{C}(A',\ A,\ B'',\ C'',\ A'_1)$ and $\mathcal{C}(A',\ S,\ C',\ B',\ A'_1)$ have the same tangent line at A' and so do the conics $\mathcal{C}(A',\ A,\ B'',\ C'',\ A'_2)$ and $\mathcal{C}(A',\ S,\ C',\ B',\ A'_2)$; these two tangent lines are the tangents of the cubic $\mathcal{T}$ at its node A'.

An analogue construction gives the tangent lines at the three nodes of the quartic $\mathcal{F}$ and at the six nodes of the quintic $\mathcal{Q}$.

3. From Hilton [2, p. 36, exer. 11], we learn that the equation of any sextic with eight given nodes is $aU^2 + 2hUV + bV^2 = CQ$, where $U = 0$, $V = 0$ are two cubics through the eight nodes, $C = 0$ is a conic through five of the nodes and $Q = 0$ is the quartic through these five nodes having double points at the other three nodes. And exercise 12 says that nine nodes of a sextic cannot be chosen arbitrarily.

From the main theorem, we know that any $\mathcal{DC}$, where each line contains exactly three points, determines a sextic with a node at any of the ten points of the $\mathcal{DC}$. Exercise 10, p. 35, says that the equation of any quintic with six given nodes 1, 2, 3, 4, 5, 6 is

$$aC_1C_2L_{12} + bC_3C_4L_{34} + cC_5C_6L_{56} = 0,$$

where $C_1 = 0$ is the conic $\mathcal{C}(2,\ 3,\ 4,\ 5,\ 6)$ and $L_{12} = 0$ is the line 12, and so on . . . .

The quintic obtained in subsection 5.1 has six nodes A, A', A'', C, C', C'' which are the vertices of two perspective triangles $AC''A'$ and $CA''C'$.

There is a remarkable sentence on page 123 of Hilton's book (although this book dates from 1932): 'Our knowledge of the theory of quintic curves (and still more of sextic) is very limited.'

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